

MTH166

Lecture-24

Solution of Wave Equation

Topic:

Solution of Partial Differential Equations

Learning Outcomes:

1. To solve one dimensional Wave Equation
2. D'Alembert's solution of infinitely long wave

Problem. Solve $\frac{\partial^2 u}{\partial t^2} = C^2 \frac{\partial^2 u}{\partial x^2}$ [1D-Wave Equation]; c^2 is Diffusivity constant.

Solution. The given one dimensional wave equation is:

$$\frac{\partial^2 u}{\partial t^2} = C^2 \frac{\partial^2 u}{\partial x^2}$$

(1) $0 \leq x \leq l$ (length), $t > 0$ (time)

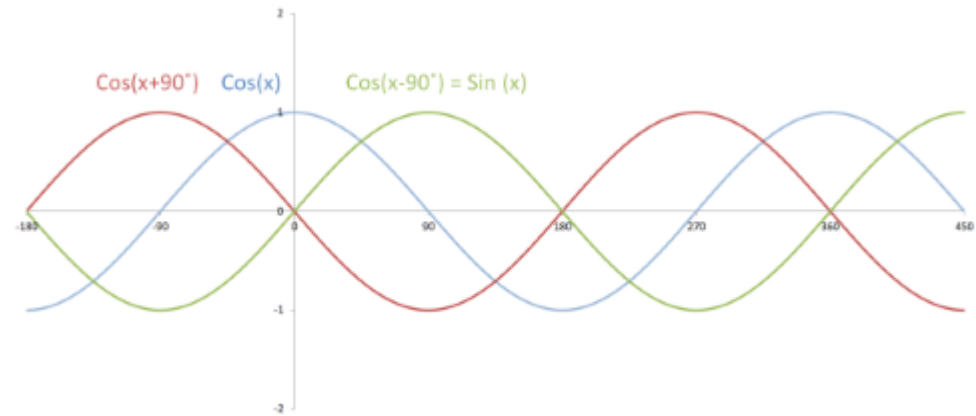
Let solution be: $u(x, t) = XT$

(2) where $X = f(x), T = g(t)$

$$\Rightarrow \frac{\partial^2 u}{\partial t^2} = XT'' \quad \text{and} \quad \frac{\partial^2 u}{\partial x^2} = X''T$$

Equation (1) becomes: $XT'' = C^2 X''T$

$$\Rightarrow \frac{X''}{X} = \frac{1}{C^2} \frac{T''}{T} = k \quad (\text{Say})$$



As k can take three values: zero, positive or negative, so we have following three cases.

Case 1. When $k = 0$

$$\frac{X''}{X} = k \quad \Rightarrow \frac{X''}{X} = 0 \quad \Rightarrow X'' = 0 \quad \Rightarrow X = ax + b$$

$$\text{Also } \frac{1}{c^2} \frac{T''}{T} = k \quad \Rightarrow \frac{1}{c^2} \frac{T''}{T} = 0 \quad \Rightarrow T'' = 0 \quad \Rightarrow T = ct + d$$

Required solution of equation (1) is:

$$u(x, t) = XT = (ax + b)(ct + d)$$

Case 2. When $k = p^2$ (Positive)

$$\frac{X''}{X} = k \quad \Rightarrow \frac{X''}{X} = p^2 \quad \Rightarrow X'' - p^2 X = 0$$

$$\text{S.F. } (D^2 - p^2)X = 0$$

$$\text{A.E. } (D^2 - p^2) = 0 \quad \Rightarrow D = \pm p$$

$$\therefore X = ae^{px} + be^{-px}$$

$$\text{Also } \frac{1}{c^2} \frac{T''}{T} = k \quad \Rightarrow \frac{1}{c^2} \frac{T''}{T} = p^2 \quad \Rightarrow T'' - C^2 p^2 T = 0$$

$$\underline{\text{S.F.}} (D^2 - C^2 p^2)T = 0$$

$$\underline{\text{A.E.}} (D^2 - C^2 p^2) = 0 \quad \Rightarrow D = \pm Cp$$

$$\therefore T = ce^{Cpt} + de^{-Cpt}$$

Required solution of equation (1) is:

$$u(x, t) = XT = (ae^{px} + be^{-px})(ce^{Cpt} + de^{-Cpt})$$

or

$$u(x, t) = XT = (a \cosh px + b \sinh px)(c \cosh Cpt + d \sinh Cpt)$$

Case 3. When $k = -p^2$ (Negative)

$$\frac{x''}{x} = k \quad \Rightarrow \frac{x''}{x} = -p^2 \quad \Rightarrow X'' + p^2 X = 0$$

$$\text{S.F. } (D^2 + p^2)X = 0$$

$$\text{A.E. } (D^2 + p^2) = 0 \quad \Rightarrow D = \pm ip$$

$$\therefore X = e^{0x}(a \cos px + b \sin px)$$

$$\text{Also } \frac{1}{C^2} \frac{T''}{T} = k \quad \Rightarrow \frac{1}{C^2} \frac{T''}{T} = -p^2 \quad \Rightarrow T'' + C^2 p^2 T = 0$$

$$\text{S.F. } (D^2 + C^2 p^2)T = 0$$

$$\text{A.E. } (D^2 + C^2 p^2) = 0 \quad \Rightarrow D = \pm iCp$$

$$\therefore T = e^{0t}(c \cos Cpt + d \sin Cpt)$$

Required solution of equation (1) is:

$$u(x, t) = XT = (a \cos px + b \sin px) (c \cos Cpt + d \sin Cpt)$$

This is the most suitable and practically feasible solution of wave equation.

Note: Wave Equation

Equation: $\frac{\partial^2 u}{\partial t^2} = C^2 \frac{\partial^2 u}{\partial x^2}$

Nature: Hyperbolic

Solution: **1.** $u(x, t) = (ax + b)(ct + d)$

2. $u(x, t) = (ae^{px} + be^{-px})(ce^{Cpt} + de^{-Cpt})$

or

$$u(x, t) = (a \cosh px + b \sinh px)(c \cosh Cpt + d \sinh Cpt)$$

3. $u(x, t) = XT = (a \cos px + b \sin px)(c \cos Cpt + d \sin Cpt)$ **(Most suitable one)**

Polling Quiz

Which of the following is a one-dimensional wave equation?

(A) $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$

(B) $\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$

(C) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

(D) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$

D'Alembert's Solution of Infinitely long wave (string)

Let given wave equation be: $\frac{\partial^2 u}{\partial t^2} = C^2 \frac{\partial^2 u}{\partial x^2}$ (1)

such that $-\infty < x < \infty, t > 0$

with initial displacement = $f(x)$ and initial velocity = $g(x)$

Then. D'Alembert's solution of equation (1) is given by:

$$u(x, t) = \frac{1}{2} [f(x + Ct) + f(x - Ct)] + \frac{1}{2C} \int_{x-Ct}^{x+Ct} g(s) ds$$

Find D' Alembert's solution of following:

Problem 1. $f(x) = \sin x, g(x) = a$

Solution. $f(x + ct) = \sin(x + ct), f(x - ct) = \sin(x - ct), g(s) = a$

D' Alembert's solution is given by:

$$u(x, t) = \frac{1}{2} [f(x + Ct) + f(x - Ct)] + \frac{1}{2C} \int_{x-Ct}^{x+Ct} g(s) ds$$

$$\Rightarrow u(x, t) = \frac{1}{2} [\sin(x + ct) + \sin(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} a ds$$

$$\Rightarrow u(x, t) = \frac{1}{2} [\sin x \cos ct + \cos x \sin ct + \sin x \cos ct - \cos x \sin ct] + \frac{a}{2c} [x + ct - x + c t]$$

$$\Rightarrow u(x, t) = \frac{1}{2} [2 \sin x \cos ct] + \frac{a}{2c} [2ct]$$

$$\Rightarrow u(x, t) = \sin x \cos ct + at \quad \textbf{Answer.}$$

Find D' Alembert's solution of following:

Problem 2. $f(x) = 0, g(x) = \cos x$

Solution. $f(x + ct) = 0, f(x - ct) = 0, g(s) = \cos s$

D' Alembert's solution is given by:

$$u(x, t) = \frac{1}{2} [f(x + ct) + f(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

$$\Rightarrow u(x, t) = \frac{1}{2} [0 + 0] + \frac{1}{2c} \int_{x-ct}^{x+ct} \cos s \, ds$$

$$\Rightarrow u(x, t) = \frac{1}{2c} [\sin s]_{x-ct}^{x+ct} = \frac{1}{2c} [\sin(x + ct) - \sin(x - ct)]$$

$$\Rightarrow u(x, t) = \frac{1}{2c} [\sin x \cos ct + \cos x \sin ct - \sin x \cos ct + \cos x \sin ct]$$

$$\Rightarrow u(x, t) = \frac{1}{c} [\cos x \sin ct] \quad \textbf{Answer.}$$

Polling Quiz

D'Alembert's solution of infinitely long wave with initial displacement $f(x)$ and initial velocity $g(x)$ is given by:

$$(A) u(x, t) = [f(x + Ct) + f(x - Ct)] + \int_{-x}^x g(s) ds$$

$$(B) u(x, t) = \frac{1}{2} [f(x + Ct) + f(x - Ct)] + \frac{1}{2C} \int_{x-Ct}^{x+Ct} g(s) ds$$

$$(C) u(x, t) = [f(x + Ct) - f(x - Ct)] + \int_{x-Ct}^{x+Ct} g(s) ds$$

$$(D) u(x, t) = \frac{1}{2} [f(x + Ct) + g(x - Ct)]$$

